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Providing Information of Objects in the Surroundings by a Handheld Device Using a Universal Pointing and Interacting Device

Abstract

This invention describes providing information of objects in the surroundings by a handheld device using a universal pointing and interacting device. The universal pointing and interacting (UPI) device is configured to identify objects in the surroundings and to interact with the handheld device such as a smartphone or a tablet. The handheld device is further configured to provide visual information on its screen or audio information by its loudspeaker or by a connected earpiece or headphones.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation-In-Part of U.S. patent application Ser. No. 18/211,313 filled on Jun. 19, 2023 which is hereby incorporated by reference in its entirety. This application claims priority benefits of U.S. patent application Ser. No. 17/524,769 filled on Nov. 12, 2021, which is hereby incorporated by reference in its entirety. This application also claims priority benefits of U.S. patent application Ser. No. 16/931,391 filled on Jul. 16, 2020, which is hereby incorporated by reference in its entirety. This application also claims priority benefits of U.S. provisional patent application Ser. No. 62/875,525 filed on Jul. 18, 2019, which is hereby incorporated by reference in its entirety. This application also claims priority benefits of U.S. provisional patent application Ser. No. 63/113,878 filed on Nov. 15, 2020, which is hereby incorporated by reference in its entirety. This application also claims priority benefits of U.S. provisional patent application Ser. No. 63/239,923 filed on Sep. 1, 2021, which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION
[0002] The present invention relates to a simple and practical universal pointing and interacting (UPI) device that can be used for interacting with objects in the surroundings of the user of the UPI device. In particular, the present invention describes providing information of objects in the surroundings by a handheld device using a universal pointing and interacting device.
BACKGROUND ART
[0003] Advances in processing and sensing technologies enable users of electronic devices to receive information about and to interact with objects in their surroundings. Applications for navigation, ride-sharing, public-transportation, commerce or similar applications on handheld devices (e.g., smart-phones or tablets) make use of the location coordinates (longitude, latitude and altitude) of the users and are ubiquitous in everyday lives. Simplified augmented reality (AR) applications on handheld devices make use of the location parameters, such as the location coordinates, the azimuth and the orientation (pitch and roll) of the handheld devices to interact with the surroundings, such as Live View navigation that superimposes information tags on images of the surroundings captured by the camera of a handheld device, or star-gazing applications that allow the user to point and identify heavenly bodies and to receive information about such bodies via the camera and the screen of a handheld device. Full scale AR systems, civilian and military, commonly use the location parameters of head-mounted displays (HMD) for user interaction with objects of interest in the surroundings, such as receiving information about or activating objects pointed at by the user's line-of-sight or by handheld control units. The core technology for the estimation of the location coordinates is satellite navigation, such as GPS, that estimates the longitude, latitude and altitude by triangulations of satellite signals (the acronym GPS is used throughout this disclosure to represent any global navigation system). The azimuth is estimated by a magnetometer that measures the earth magnetic field and the orientation (pitch and roll) is estimated by an accelerometer that measures the earth gravity vector. The accuracy of the estimation of the location parameters may improve in urban and indoor environments by triangulating wireless signals such as cellular, Wi-Fi, Bluetooth, or dedicated radio-beacons. Further accuracy in the estimation of the location parameters may be achieved by analyzing images of the surroundings taken by the camera of a handheld device for the detection of objects of interest with known pictorial representations and known locations. Once such objects of interest are detected, their known locations and the positions they appear in the camera field-of-view may be used for very accurate triangulations to establish the location parameters of the handheld device. This approach became feasible in recent years as visual descriptions and location parameters data of vast numbers of topographical, urban and commercial objects of interest are now available in

accessible databases, as the result of extensive photographic and geographic surveys that were carried out at all corners of the world during the last decade.

[0004] The following US patents and patent applications disclose several background art aspects that are relevant to this invention, such as approaches of location parameters estimation, motion detection, styluses and wands, pointing and activating devices and AR systems, image assisted orientation and navigation, as wells as other aspects. U.S. Pat. Nos. 6,897,854, 7,596,767, 8,179,563, 8,239,130, 8,467,674, 8,577,604, 8,681,178, 8,698,843, 8,810,599, 8,878,775, 9,140,555, 9,195,872, 9,201,568, 9,224,064, 9,229,540, 9,280,272, 9,303,999, 9,329,703, 9,400,570, 9,429,434, 9,519,361, 9,639,178, 9,678,102, 9,696,428, 9,870,119, 10,028,089, 10,168,173, 10,198,649, 10,198,874, 10,240,932, 10,274,346, 10,275,047, 10,275,051, 10,281,994, 10,318,034, 10,324,601, 10,328,342, 10,365,732, 10,444,834, 10,488,950, 10,521,028, 10,565,724, 10,5920,10, and US patent applications 2018/0173397, 2019/0369754, 2020/0103963.

[0005] Immersive AR systems that use HMD may be able to provide a full AR experience, including the interaction with real and virtual objects in the environment, but are expensive, difficult to operate and were not yet shown to attract significant commercial attention. Simplified AR applications on handheld devices may have wider commercial penetration, but are quite uncomfortable and awkward to operate, as pointing at objects of interest is not intuitive and as the screen of a handheld device is used for both viewing the surroundings and entering user commands to interact with objects of interest in the surroundings. Moreover, simplified AR navigation applications on handheld devices can pose a danger to the user who holds the handheld device at eye level, causing significant distraction from the immediate surroundings. Therefore, there is a need for a cheap and simple device that can provide comfortable and intuitive pointing and interacting capabilities with objects of interest in the user's surroundings.

BRIEF DESCRIPTION OF THE INVENTION

[0006] The present invention is directed to a universal pointing and interacting (UPI) device that is configured to interact with objects of interest, where interacting means identifying objects of interest pointed at by the UPI device and at least one of providing information about such objects and activating such objects by the user of the UPI device. Objects of interest are geographical, topographical, urban, commercial and other types of objects in the surroundings that their features are readily tabulated in vast and accessible features databases. The features of the objects of interest are at least the location parameters of the objects of interest, but for most objects of interest the features may include visual descriptions of the objects of interest, such as the objects of interest pictorial representation or structural data. The features of the objects of interest may also include any additional useful information about such objects. The term “features database” is used throughout this disclosure to denote any publicly or privately available database that may be used to retrieve these features of the objects of interest. Objects of interest may also be moving objects, such as public transportation vehicles, that estimate their location coordinates and that make their location coordinates available over the Internet or other networks. Objects of interest may also by virtual objects that are generated with known locations in the surroundings of the user of the UPI device. A particular type of objects of interest are display screens, where the UPI device may be used to control such screens, similar to a mouse, remote control or stylus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The features and advantages of the present invention will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, wherein:

[0008] FIG. **1**A is a schematic diagram schematic diagram of a full and independent UPI device.

[0009] FIG. **1**B is a schematic diagram schematic diagram of a simplified UPI Device.
[0010] FIG. **2**A is a schematic diagram an operational configuration for a UPI device.
[0011] FIG. **2**B is a schematic diagram of an operational configuration for a simplified UPI device.
[0012] FIG. **3** is a schematic flowchart of a scanning procedure, with user actions on the left and UPI device actions on the right.
[0013] FIG. **4** is a schematic flowchart of a locating procedure, with user actions on the left and UPI device actions on the right.
[0014] FIG. **5** is a schematic flowchart of a navigating procedure, with user actions on the left and UPI device actions on the right.
[0015] FIG. **6** is a schematic diagram of examples of screens pointed at and activated by a UPI device.
[0016] FIG. **7** is a schematic diagram of the operation of a UPI device in pointing and activating screens.
[0017] FIG. **8** is a schematic diagram of examples of interaction by UPI device with real and virtual objects on a screen of a handheld device.

DETAILED DESCRIPTION OF THE INVENTION

[0018] FIG. **1**A describes a full and independent universal pointing and interacting (UPI) device **100**. The components of UPI device **100** in FIG. **1**A correspond to the components of the UPI device described in FIG. **1** of U.S. patent application Ser. No. 16/931,391. Such UPI device **100** comprises of an elongated body similar to a wand or a stylus. Tip **102** of UPI device **100** is the frontal end of the longest dimension of the elongated body of UPI device **100** and the main axis of UPI device **100** is the axis of longest dimension of the elongated body of UPI device **100**. Pointing ray **105** is an indefinite extension of the main axis of UPI device **100** in the frontal direction from tip **102** of UPI device **100**. Camera **110** and LIDAR+ **115** are positioned at tip **102** of UPI device **100**. LIDAR+ **115** is any type of LIDAR device, single-beam or multi-beam, or it can be any other distance measuring device that may be based on other technologies, such as ultrasound or radar. UPI device **100** may also be of any shape, other than an elongated body, that is suitable to be held by hand, clipped on eyeglasses frames, strapped to the user head (e.g., attached to a hat, visor or headband), attached to any part of the user body, or shaped as part of a ring, where the frontal direction of UPI device **100** that forms pointing ray **105** is the direction the camera of UPI device **100** is facing. UPI device **100** further includes location sensing components **120**, user input components **130**, user output components **140**, control components **150**, and battery **160**. Location sensing components **120** includes accelerometer **122**, magnetometer **124** and gyroscope **126**, as well as GPS **128**. User input components **130** includes tactile sensors **132** (e.g., switches, slides, dials or other touch sensors), microphone **134** and fingerprint detection sensor **136**. User output components **140** includes light emitting diode (LED) **142**, vibration motor **144**, loudspeaker **146** and screen **148**. Control components **150** includes computation component **152** and communication component **154**.
[0019] Input components **130** and output components **140** facilitate the user interaction with UPI device **100**. Specifically, if UPI device **100** is designed to be held by a hand, it may have one side that is mostly used facing up and an opposite side that is mostly used facing down. Tactile sensors **132** and fingerprint detection sensor **136** are placed on the outer shell of UPI device **100** in suitable locations that are easily reachable by the fingers of the user, for example, at the “down” side. LED **142** (or several LEDs) are also placed on the outer shell of UPI device **100** in suitable locations to be seen by the user, for example, at the “up” side. Screen **148** may also be placed on the outer shell of UPI device **100** at the “up” side. Vibration motor **144** is placed inside UPI device **100**, preferably close to the area of UPI device **100** where the user is holding the device. Moreover, two units of vibration motor **144** may be used, each placed in each edge of UPI device **100**, which can be used to create rich vibration patterns for the user of UPI device **100**. Microphone **134** and loudspeaker **146** are placed for optimal receiving of audio and playing of audio from/to the user, respectively.

[0020] As was discussed in U.S. patent application Ser. No. 16/931,391, UPI device **100** depicted in FIG. **1**A can operate as an independent device. However, as UPI device **100** is more likely to operate together with a handheld device, such as a cellphone or a tablet, some of the components of UPI device **100** are not essential. For example, GPS **128** may not be essential, as the general location of UPI device **100** may be obtained from the location information of the handheld device if UPI device **100** is operating together with that handheld device. FIG. **1**B describes a simplified option for the structure of UPI device **100**, which includes the key elements of camera **110** and control components **150** that transmit the images captured by camera **110** to the handheld device. Even battery **160** may be eliminated if UPI device **100** is connected by a cable to the handheld device. UPI device **100** can have any configuration between the full and independent configuration depicted in FIG. **1**A and the simplified configuration depicted in FIG. **1**B. Obviously, in any configuration that is not the full and independent configuration depicted in FIG. **1**A, UPI device **100** will be operating together with a handheld device and therefore the description of the operation of UPI device **100** may also be considered as a description of the operation of UPI device **100** that is operating together with that handheld device, as discussed below.
[0021] FIG. **2**A shows an optional operational configuration for UPI device **100** where user **200** of UPI device **100** also uses handheld device **205** (e.g., smart-phone, tablet, etc.) and optionally also earbuds **210** that may include a microphone. Wireless connections **215**, **220** and **225** connect UPI device **100**, handheld device **205** and earbuds **210**, based on the desired configuration, and are commonly Bluetooth or Wi-Fi connections, but any other wireless or wireline connection protocol may be used. Wireless connections **215**, **220** and **225** enable the shared operation of UPI device **100** together with handheld device **205** and earbuds **210**. One element of the shared operation is the user interaction, by receiving inputs from user **200** by input elements on handheld device **205** or earbuds **210** (in addition to receiving inputs by user input components **130** on UPI device **100**) and by providing outputs to user **200** by output elements on handheld device **205** or earbuds **210** (in addition to providing outputs by user output components **140** on UPI device **100**). Another element of the shared operation is the sharing of measurements, such as using the GPS information of handheld device **205** for the operation of UPI device **100**, while yet another element is the sharing of the computation loads between UPI device **100** and handheld device **205**. Communication link **240** provides the means for UPI device **100** or handheld device **205** to connect with features database **250** that holds the features of objects of interest **230**, where the features are the location information of objects of interest **230**, visual description of objects of interest **230**, as well as other information about objects of interest **230** (e.g., opening hours and a menu if a particular object of interest **230** is a restaurant). Wireless connections **215**, **220** and **225** depicted in FIG. **2**A may be implemented using a wire connection, as earbuds **210** may be replaced with headphones connected by a wire to handheld device **205** or a wire may be used to connect between UPI device **100** and handheld device **205**.
[0022] U.S. patent application Ser. No. 16/931,391 describes the identification of objects of interest **230** that are pointed at by UPI device **100**, such that information about these objects is provided to the user. This identification of objects of interest **230** and providing that information is also critical for the blind or visually impaired, but additional operating procedures of UPI device **100** for the blind or visually impaired are the scanning of the surroundings, the locating of targets in the surroundings, as well as the navigating in the surroundings to a specific destination. To perform these operating procedures, it is critical to know the exact location of the UPI device **100** and its exact pointing direction, i.e., its azimuth and orientation. Current navigation devices (or apps on handheld devices) mainly use the GPS location information, but the error in the GPS location information is a few meters in typical situations and the error can be significantly larger in a dense urban environment. Current navigation devices may use a magnetometer to estimate the azimuth, but a typical magnetometer error is about 5° and the error can be significantly larger when the magnetometer is near iron objects. Obviously, these accuracies are insufficient for the guidance of

the blind or visually impaired uses.

[0023] The discussion of the operation of UPI device **100** in U.S. patent application Ser. No. 16/931,391 includes the description of a procedure that finds the exact location, azimuth and orientation of UPI device **100** by capturing images by camera **110**. This procedure includes identifying several objects of interest **230** in the surroundings that their locations and visual descriptions are known and tabulated in features database **250** and obtaining highly accurate estimation of the location, azimuth and orientation of UPI device **100** by triangulation from the known locations of these identified objects of interest **230**. This visual positioning procedure is identical to currently available commercial software, and specifically to the Visual Positioning Service/System (VPS) developed by Google. To avoid confusion and to align this patent application with currently published technical literature, it is important to emphasis that objects of interest **230** in U.S. patent application Ser. No. 16/931,391 comprise of two types of objects. One type of objects of interest **230** are objects that are needed for the visual position procedure, which are of very small dimension (e.g., a specific window corner) and are usually called “reference points” in the literature. The features of these reference points that are stored in features database **250** include mostly their locations and their visual description. Other type of objects of interest **230** are in general larger objects (e.g., a commercial building) and their features that are stored in features database **250** may include much richer information (e.g., commercial businesses inside building, opening hours, etc.).

[0024] In general terms, there are 3 procedures that are performed by a seeing person for orientation and navigation in the surroundings. The first procedure is “scanning”, which is performed when a person arrives to a new location, which happens, for example, when a person turns a street corner, exits a building or disembarks a vehicle. The second procedure is “locating”, which is performed when a person is interested in locating a particular object or landmark in the surroundings, such as a street address, an ATM or a bus stop. The third procedure is “navigating”, which is the controlled walking from a starting point to a destination end point. To perform these procedures, a seeing person is using visual information to determine the person location and orientation. Obviously, a seeing person seamlessly and interchangeably uses these 3 procedures in everyday activities. We will describe how a blind or visually impaired person can use UPI device **100** to perform each of these procedures.

[0025] The scanning procedure is performed by holding UPI device **100** and moving it along an approximated horizontal arc that covers a sector of the surroundings. As UPI device **100** is estimating its location and as it is moved along an approximated horizontal arc its azimuth and orientation are continuously updated and therefore it can identify objects of interest **230** in its forward direction as it is moved. UPI device **100** can provide audio descriptions to the user about objects of interest **230** at the time they are pointed at by UPI device **100** (or sufficiently close to be pointed at), based on data obtained from features database **250**. The scanning procedure is initiated by touch or voice commands issued when the user wants to start the scanning, or simply by self-detecting that UPI device **100** is held at an approximated horizontal position and then moved in an approximated horizontal arc. Moreover, as UPI device **100** is fully aware of its location, UPI device **100** can also be configured to alert the user about a change in the surroundings, such as rounding a corner, to promote the user to initiate a new scanning process.

[0026] A typical urban environment contains numerous objects of interest and a seeing person who is visually scanning the surroundings may instinctively notice specific objects of interest at suitable distances and of suitable distinctions to obtain the desired awareness of the surroundings. While it is difficult to fully mimic such intuitive selection by UPI device **100**, several heuristic rules may be employed for an efficient and useful scanning procedure for the blind or visually impaired users. One rule can be that audio information should be provided primarily about objects that are in line-of-sight of UPI device **100** and that are relatively close. Another rule may limit the audio information to objects that are more prominent or more important in the surroundings, such as playing rich audio information about a building of commercial importance pointed at by UPI device **100**, while avoiding playing information about other less significant buildings on a street. Yet another rule can be the control of the length and depth of the audio information based on the rate of motion of UPI device **100** along the approximated horizontal arc, such that the user may receive less or more audio information by a faster or slower horizontal motion of UPI device **100**, respectively. Obviously, the parameters of these rules may be selected, set and adjusted by the user of UPI device **100**. The audio information can be played by earbuds **210**, loudspeaker **146** or the loudspeaker of handheld device **205**, and can be accompanied by haptic outputs from vibration motor **144**, or by visual outputs on screen **148** or on the screen of handheld device **205**. The presentation of the audio information can be controlled by the motion of UPI device **100**, by touch inputs on tactile sensors **132**, or by voice commands captured by a microphone on earbuds **210**, microphone **134** or the microphone of handheld device **205**. For example, a short vibration may indicate that UPI device **100** points to an important object of interest and the user may be able to start the playing of audio information about that object by holding UPI device **100** steady, touching a sensor or speaking a command. Similarly, the user can skip or stop the playing of audio information by moving UPI device **100** further, touching another sensor or speaking another command.

[0027] FIG. **3** is a schematic flowchart of the scanning procedure performed by a blind or visually impaired person using UPI device **100**. The actions taken by the user of UPI device **100** are on the left side of FIG. **3**, in dashed frames, while the actions taken by UPI device **100** are on the right side of FIG. **3**, in solid frames. In step **300**, the user lifts UPI device **100** to horizontal position and issues a command to UPI **100** to start a scanning procedure. The issuing of the command may be explicit, such as by any of tactile sensors **132** or by a voice command, or may be implicit by holding UPI device **100** stable in a horizontal position, which may be recognized by UPI device **100** as the issuing of the scan command. UPI device **100** receives or identifies the scan command and estimates its exact location, azimuth and orientation in step **310**. The exact estimation procedure was described in U.S. patent application Ser. No. 16/931,391 and it uses the initial location information obtained by GPS signals, the initial azimuth generated from measurements by magnetometer **124**, the initial orientation generated from measurements by accelerometer **122** and a frontal visual representation generated by processing an image captured by camera **110**, to identify a set of objects of interest **230** ("reference points") in the surroundings. The identification is performed by a comparison (using correlation) of the content of the frontal visual representation with the visual description of objects of interest **230** in the surroundings. Obviously, the initial location information is the key parameter that helps to narrow performing the comparison process only for objects of interest **230** that are in the surroundings, while not performing the comparison process for objects of interest **230** elsewhere. The initial azimuth and the initial orientation further assist in narrowing the performing the comparison process only for objects of interest **230** that are in a specific sector of the surroundings, while not performing the comparison process for objects of interest that are not in that sector. Once such set of objects of interest **230** is identified, the locations of these identified objects of interest **230** are obtained from database **250** and the exact location, azimuth and orientation of UPI device **100** is estimated using triangulation of the obtained locations of these identified objects of interest **230**. In step **320** UPI device **100** identifies further objects of interest **230** in the surroundings as further objects of interest **230** that are pointed at pointing ray **105**, i.e., objects of interest **230** that are in line-of-sight and that pointing ray **105** passes within a predetermined distance to them. Note that the further identified objects of interest **230** may be different than the set of objects of interest **230** ("reference points") that were identified for the estimation of the exact location, azimuth and orientation of UPI device **100** in step **310**. UPI device **100** then provides information about the further identified objects of interest **230** to the user in step **330**. As the user moves UPI device **100** in horizontal direction to scan the surroundings in step **340**, UPI device **100** continuously further estimates its location, azimuth and orientation and repeats steps **320** and **330**.

[0028] The following three examples demonstrate some superior aspects of scanning the surroundings by UPI device **100** over visual scanning of the surroundings by a seeing person. In the first example, UPI device **100** may also provide audio information about objects of significant importance in the surroundings, such as a mall, a landmark, a transportation hub or a house of warship, that might be very close but not in the direct line-of-sight of UPI device **100** (e.g., just around a corner). In the second example, the audio information about the pointed-at objects of interest **230** may include details that are not directly visible to a seeing person, such as menus of restaurants, lists of shops and offices inside a commercial building, or operating hours of a bank. In the third example it is assumed that UPI device **100** is pointed to a fixed alphanumeric information in the surroundings (such as street signs, stores and building names, informative or commemoration plaques, etc.). As the locations and the contents of most alphanumeric information in the public domain are likely to be tabulated in and available from features database **250**, the alphanumeric information may be retrieve from features database **250**, converted to an audio format and provided to the user of UPI device **100** regardless of the distance, the lighting or the viewing-angle of the alphanumeric information. The information about the object in the surroundings may also by provided by the screen of handheld device **205** in alphanumerical format or image format, or converted to audio and the audio by handheld device **205** played by the loudspeaker of handheld device **205** or by earbuds **210** or any other type of earpiece or headphones.

[0029] A seeing person may intuitively locate a specific target in the surroundings, such as an ATM, a store or a bus stop. Blind or visually impaired users of UPI device **100** are able to initiate a locating procedure for particular targets, such as “nearest ATM”, using a touch or voice-activated app on handheld device **205**, or, alternatively, by a touch combination of tactile sensors **132** on UPI device **100**. The app or UPI device **100** may then provide the blind or visually impaired user with information about the target, such as the distance to and the general direction of the target, or any other information about the target such as operating hours if the target is a store, an office or a restaurant. If the user of UPI device **100** is only interested in reaching that specific target, the next step is to activate a navigating procedure, which is described further below, and to start walking toward the target. It is possible, however, that the user of UPI device **100** may want to know the general direction of a specific target or the general directions of several targets to be able to choose between different targets. To get an indication of the general direction of a specific target the user may lift UPI device **100** and point it to any direction, which will allow UPI device **100** to obtain current and accurate estimates of its location, azimuth and orientation. Once these estimates are obtained, UPI device **100** may use several method to instruct the user to manipulate the pointing direction of UPI device **100** toward the target, such as using audio prompts as “quarter circle to your left and slightly upward”, playing varying tones to indicate closeness or deviation from the desired direction, or using vibration patterns to indicate closeness or deviation from the desired direction.

[0030] FIG. **4** is a schematic flowchart of the locating procedure performed by a blind or visually impaired person using UPI device **100**. The actions taken by the user of UPI device **100** are on the left side of FIG. **4**, in dashed frames, while the actions taken by UPI device **100** are on the right side of FIG. **4**, in solid frames. In step **400**, the user provides the target information, such as “City Museum of Art” or “nearest ATM”. In step **410**, UPI device **100** finds the target information in features database **250** of objects of interest **230**. Note that UPI device **100** always has the general information about its location, using its GPS device **128** or the GPS data of handheld device **200**, which allows it to find distances relative to its location, such as the nearest ATM. As the target information is found, in step **420** the user lifts UPI device **100** to horizontal position. In step **430** UPI device **100** estimates its exact location, azimuth and orientation, as described above in the discussion of FIG. **3**. Once the exact location, azimuth and orientation of UPI device **100** are estimated, in step **440** UPI device **100** provides motion instructions to the user on how to move UPI device **100** such that UPI device **100** is pointing closer and closer to the direction of the target. In step **450** the user moves UPI device **100** according to the instructions. UPI device **100** then repeats the estimates its exact location, azimuth and orientation and provides further motion instructions to the user in steps **430** and **440**, respectively. Obviously, when UPI **100** points directly to the target, UPI device **100** informs the user of the successful completion of the locating procedure.

[0031] Targets may be stationary targets, but can also be moving targets that their locations are known, such as vehicles that make their locations available (e.g., buses, taxies or pay-ride vehicles) or people that carry handheld devices and that consensually make their locations known. For example, a seeing person may order a pay-ride vehicle using an app and will follow the vehicle location on the app's map until the vehicle is close enough to be seen, where at that point the seeing person will try to visually locate and identify the vehicle (as the make, color and license plate information of pay-ride vehicles are usually provided by the app). As the vehicle is recognized and is approaching, the seeing person may raise a hand to signal to the driver and might move closer to the edge of the road. A blind or visually impaired person may be able to order a pay-ride vehicle by voice activating a reservation app and may be provided with audio information about the progress of the vehicle, but will not be able to visually identify the approaching vehicle. However, using UPI deice **100**, as the location of UPI device **100** is known exactly and as the location of the pay-ride vehicle is known to the app, once the vehicle is sufficiently close to be seen, UPI device **100** may prompt the user to point it in the direction of the approaching vehicle such that an image of the approaching vehicle is captured by camera **110**. The image of the approaching vehicle may then be analyzed to identify the vehicle and audio information (such as tones) may be used to help the user in pointing UPI device **100** at the approaching vehicle, such that updated and accurate information about the approaching vehicle may be provided to the user of UPI device **100**. Obviously, the same identifying and information providing may be used for buses, trams, trains and any other vehicle with a known position. In yet another example, a seeing person may be able to visually spot a friend at some distance on the street and to approach that friend, which is impossible or difficult for a blind or visually impaired person. However, assuming that friends of a blind or visually impaired person allow their locations to be known using a special app, once such a friend is sufficiently close to the user of UPI device **100**, the user of UPI device **100** may be informed about the nearby friend and the user may be further assisted in aiming UPI device **100** in the general direction of that friend.

[0032] UPI device **100** can also operate as a navigation device to guide a blind or visually impaired user in a safe and efficient walking path from a starting point to a destination point. Walking navigation to a destination is an intuitive task for a seeing person, by seeing and choosing the destination target, deciding on a path to the destination and walking to the destination while avoiding obstacles and hazards on the way. Common navigation apps in handheld devices may assist a seeing person in identifying a walking destination that is further and not in line-of-sight, by plotting a path to that destination and by providing path instructions as the person walks, where a seeing person is able to repeatedly and easily compensate and correct the common but slight errors in the navigation instructions. Assisted navigation for blind or visually impaired users of UPI device **100** is a complex procedure of consecutive steps that need to be executed to allow accurate and safe navigation from the location of the user to the destination. This procedure differs from the navigation by a seeing person who is helped by a common navigation app on a handheld device. Unlike the very general walking directions provided by a common navigation app, assisted navigation for the blind or visually impaired needs to establish a safe and optimal walking path tailored to the needs of the blind or visually impaired, and to provide precise guidance through the established walking path, while detecting and navigating around obstacles and hazards.

[0033] The navigating procedure starts with the choice of a walking destination using a navigation app, which may be performed by a blind or visually impaired person using voice commands or touch commands, as described above for the locating procedure. Once the walking destination and its location are established, the location of the user needs to be determined. An approximated location of the user may be obtained using GPS signals, but a more accurate location can be established by pointing UPI device **100** to any direction in the surroundings to obtain an estimation of the location by visual triangulations. (As some pointing directions may provide more reference points for more accurate visual triangulation, UPI device **100** may use voice prompts, tones or vibrations to instruct the user to point toward an optimal direction for improved estimation of the user location.) UPI device **100** may then inform the user about the accuracy of the estimation such that the user is aware of the expected accuracy of the navigation instructions. Once a sufficiently accurate (or best available) estimation of the user location is obtained, a navigation route from the user location to the location of the walking destination is planned and calculated. A specific route for blind or visually impaired users should avoid or minimize obstacles and hazards on the route, such as avoiding steps, construction areas or narrow passages, or minimizing the number of street crossings. The specific route should steer the user of UPI device **100** away from fixed obstacles, such as lampposts, street benches or trees, where the data about such fixed obstacles may be obtained from features database **250**. Further, current visual data from CCTV cameras may show temporary obstacles, such as tables placed outside of a restaurant, water puddles after the rain or a gathering of people, and that visual data may be used to steer the user of UPI device **100** away from these temporary obstacles. In addition to considering the safety and the comfort of the blind or visually impaired user of UPI device **100**, the route planning may also take into account the number and the density of reference points for visual triangulations along the planned walking route, such that the estimation of the user location and direction can be performed with sufficient accuracy throughout the walking route.

[0034] A seeing person may simply walk along the navigation route and use visual cues for directional orientation and for following the route, which is not possible for a blind or visually impaired person. Instead, the pointed structure UPI device **100** (e.g., its elongated body) may be used to indicate the walking direction for the blind or visually impaired users of UPI device **100**. To start the walk, the user may hold UPI device **100** horizontally at any initial direction and UPI device **100** will then provide voice prompts, varying tones or vibrating patterns to guide the user in pointing UPI device **100** to the correct walking direction, as described above for the locating procedure. As the user walks, voice prompts, varying tones or vibrating patterns (or combination of these) may be continuously used to provide walking instructions, warnings of hazards and turns, guiding the correct position of UPI device **100**, or providing any other information that makes the navigation easier and safer. UPI device **100** can use the data in features database **250** to safely navigate the user around fixed obstacles, but it may also use the information from camera **110** or LIDAR+ **115** to detect temporary obstacles, such as a trash bin left on the sidewalk or a person standing on the sidewalk, and to steer the user of UPI device **100** around these obstacles.

[0035] FIG. **5** is a schematic flowchart of the navigating procedure performed by a blind person using UPI device **100**. The actions taken by the user of UPI device **100** are on the left side of FIG. **5**, in dashed frames, while the actions taken by UPI device **100** are on the right side of FIG. **5**, in solid frames. In step **500**, the user specifies the walking destination, such as “Main Mall” or “nearest Italian restaurant”. In step **510**, UPI device **100** finds the walking destination information in database **250**. As the walking destination information is found, at step **520** the user lifts UPI device **100** to horizontal position. At step **530** UPI device **100** estimates its exact location, azimuth and orientation (as described above in the discussion of FIG. **3**), and calculates or refines a walking route from the location of the user of UPI device **100** to the walking destination. Once the exact location, azimuth and orientation of UPI device **100** are estimated and the walking route is calculated or refined, UPI device **100** provides motion instructions to the user on how to move UPI device such that UPI device **100** will point closer and closer toward the walking direction in step **540**. As the user moves UPI device **100** according to the instructions in step **550**, UPI device **100** repeats the estimates of its location, azimuth and orientation and providing further motion instructions to the user in steps **530** and **540**, respectively. Once UPI device **100** points directly to the walking direction, it can inform the user that it is now pointing to the correct walking direction. As UPI device **100** now points to the correct walking direction, at step **560** UPI device **100** provides the user with walking instructions, such as “walk forward **20** meters”, which the user performs in step **570**. As the user walks, UPI device **100** repeats the estimates its location, azimuth and orientation and the refinement of the walking route in step **530** and, as needed, steps **540**, **550**, **560** and **570** are repeated.

[0036] Scene analysis is an advanced technology of detecting and identifying objects in the surroundings and is already employed in commercial visual substitution products for the blind or visually impaired. Scene analysis algorithms use images from a forward-facing camera (usually attached to eyeglasses frames or head-mounted) and provide information describing the characteristics of the scene captured by the camera. For example, scene analysis may be able to identify whether an object in front of the camera is a lamppost, a bench or a person, or whether the path forward is smoothly paved, rough gravel or a flooded sidewalk. Scene analysis employs features extraction and probability-based comparison analysis, which is mostly based on machine learning from examples (also called artificial intelligence). Unfortunately, scene analysis algorithms are still prone to significant errors and therefore the accuracy of scene analysis may greatly benefit from the precise knowledge of the camera location coordinates and the camera angle. Using the camera location and angle may allow a more precise usage of the visual information captured by the camera in the scene analysis algorithms. In one example, fixed objects in the surroundings can be analyzed or identified beforehand, such that their descriptions and functionalities are stored in features database **250**, which may save computation from the scene analysis algorithms or increase the probability of correct analysis of other objects. In another example of identifying whether an object in front of the camera is a lamppost, a bench or a person, a scene analysis algorithm may use the known exact locations of the lamppost and the bench in order to improve the identification that a person is leaning on the lamppost or is sitting on the bench. In yet another example, if it is known that camera **110** of UPI device **100** is pointing toward the location of sidewalk water drain, the probability of correctly detecting a water puddle accumulated during a recent rain may be greatly improved, such that the navigation software may steer the blind or visually impaired user away from that water puddle. Moreover, using the pre-captured visual and geographical information in features database **250**, possibly with the multiple current images captured by camera **110** of the walking path in front of the user as the user walks forward, a 3D model of the walking path may be generated and the user may be steered away from uneven path or from small fixed or temporary obstacles on the path.

[0037] An interesting and detailed example of combining information from several sources is the crossing of a street at a zebra-crossing with pedestrian traffic lights. Using the accurate location estimation, UPI device **100** may lead the blind or visually impaired user toward the crossing and will notify the user about the crossing ahead. Moreover, the instruction from UPI device **100** may further include details about the crossing structure, such as the width of the road at the crossing, the expected red and green periods of the crossing traffic light, the traffic situation and directions, or even the height of the step from the sidewalk to the road. Pedestrian traffic lights may be equipped with sound assistance for the blind or visually impaired, but regardless of whether such equipment is used, UPI device **100** may direct the user to point it toward the pedestrian crossing traffic lights and may be configured to identify whether the lights are red or green and to notify the user about the identified color. Alternatively, the color of the traffic lights may be transmitted to UPI device **100**. Once the crossing traffic lights change from red to green, UPI device **100** may inform the user about the lights change and the user may then point UPI device **100** toward the direction car traffic approaches the crossing. The image captured by camera **110** and the measurements by LIDAR+ **115** may then be analyzed to detect whether cars are stopped at the crossing, no cars are approaching the crossing or safely decelerating cars are approaching the crossing, such that it is safe for the user to start walking into the crossing. On the other hand, if that analysis detects that a car is moving unsafely toward the crossing, the user will be warned not to walk into the crossing. Traffic information may also be transmitted to and utilized by UPI device **100** from CCTV cameras that are commonly installed in many major street junctions. As the user crosses the road, UPI device **100** may inform the user about the progress, such as the distance or the time left to complete the crossing of the junction. In a two-ways road, once the user reaches the center of the junction, UPI device **100** may indicate the user to point it to the other direction to be able to analyze the car traffic arriving from that direction. As the user reaches the end of the crossing, UPI device **100** may indicate the completion of the crossing and may provide the user with additional information, such as the height of the step from the road to the sidewalk.

[0038] The usage of camera **110** was described above in visual triangulations to obtain exact estimations of the location, azimuth and ordination of UPI device **100** and in scene analysis to better identified and avoid obstacles and hazards and to provide richer information about the surroundings. Further, similar to currently available products for the blind or visually impaired that include forward-looking camera, camera **110** may also be used to allow the blind or visually impaired users of UPI device **100** to share an image or a video with a seeing person (e.g., a friend or a service person) who can provide help to the users of UPI device **100** by explaining an unexpected issue, such as roadblocks, constructions or people gathering. Moreover, in addition to the image or the video, UPI device **100** may also provide the seeing person with its exact location and the direction it points to, which can be used by the seeing person to get a much better understanding of the unexpected issue by using additional resources, such as emergency service resources or viewing CCTV feeds from the exact surroundings of the user of UPI device **100**.

[0039] Several operation methods were described above in using UPI device **100** for visual substitution for the blind or visually impaired. It was shown that the unique information gathering and connectivity of UPI device **100** may be able to provide the blind or visually impaired with information that is not available for a seeing person, such as operating hours of businesses, reading of signs without the need to be in front of the signs, or noticing a nearby friend. Obviously, seeing people may also benefit for these features of UPI device **100**. Moreover, several other functions may be performed by UPI device **100** for the benefit of seeing people. Most interestingly, UPI device **100** may be used as an optical stylus, as discussed extensively in U.S. patent application Ser. No. 16/931,391. In another example, the image or video captured by camera **110** may be displayed on the screen of handheld device **205** and used for inspecting narrow spaces or to capture a selfie image. Further, video calls using the front-facing camera of handheld device **205** (or a webcam of a laptop computer) are extensively used for personal and business communications, as well as for remote learning. During such video calls it is common to want to show an object that is outside the field view of the camera used for the call, such as drawings on a book page or on a whiteboard, or a completed handwritten mathematical exercise. In such cases, instead of bothering to move the object to the field view of the camera used for the video call (e.g. the camera of handled device **205**), the user of UPI device **100** can simply aim camera **110** of UPI device **100** toward the object, such that camera **110** may capture an image or video feed of that object, which can then be sent to the other side of the video call. The video feed from camera **110** can replace the video feed from the front-facing camera of handheld device **205** (or the video feed from a webcam of a laptop computer) or both video feeds may be combined using a common picture-in-picture approach.

[0040] FIG. **6** provides examples of display screens that are used for interaction with computerized equipment, such as computer screen **610**, TV screen **620**, touchscreen in handheld device **630** (which might be the same as handheld device **205** or a different handheld device) or presentation screen **640** in a classroom or a lecture hall. Such screens are used to display information that is generated by the computerized equipment for the users, but are also used as tools for entering user input with keyboards or with common pointing elements such as mouse, remote control, fingers or stylus. UPI device **100** may be used for pointing at such screens, identifying a pointed at screen and activating the identified screen, in a method that is particular to screens, but is comparable to the method of pointing, identifying and activating objects of interest presented in FIGS. **3** and **6** and discussed above. A first difference is that each screen renders visual information generated by the computerized equipment, which means that as UPI device **100** points at a screen, the at least one image captured by camera **110** contains not only the outline of that screen, but also the visual information rendered on that screen. A representation of the content rendered on the screens (which we call “screen content data”) may be transmitted from the computerized equipment to handheld device **205** or UPI device **100**. In addition, visual descriptions of outlines of screens may be obtained from private or public features database **250**. Therefore, visual descriptions of screens may be generated using at least one of the screens content data and the obtained visual descriptions of outlines of screens. The generated visual descriptions of screens may be used for the identification of a particular screen, similar to using stored visual descriptions of objects of interest from public or private features databases for the identification of a particular object of interest, as described in details by U.S. patent application Ser. No. 16/931,391. A second difference is that the activating of the identified screen by UPI device **100** may be a continuous process of activating and modifying the content displayed on the screen in the same way that common pointing elements are used for activation and control of display screens.

[0041] FIG. **7** is a schematic flowchart of an operation method of UPI device **100** in pointing at display screens, identifying a pointed at screen and activating the identified screen based on the relative position between UPI device **100** and the identified screen and based on user **200** input on UPI device **100**. This operation method was described in detail by U.S. Patent application Ser. No. 16/931,391. The identification of a screen may be considered as identifying a particular screen from candidates' screens, but it can also be considered as confirming that UPI device **100** is properly positioned and points to a particular screen, ready to activate that screen. Steps **710-740** in FIG. **7** are parallel to steps **310-340** in FIG. **3** of U.S. patent application Ser. No. 16/931,391, where the only difference is that in step **740** the transmitting of the estimated orientation of UPI device **100**, the estimated azimuth of UPI device **100** and the generated frontal visual representation for UPI device **100** by communication component **154** from UPI device **100** to handheld device **205** is for the identification of a screen pointed at by UPI device **100**. Step **750** describes obtaining features of candidates' screens from features database **250**, wherein the features of candidates screens comprise of at least one of the location parameters of candidates' screens and the visual descriptions of outlines of candidates screens, receiving screen content data of candidates' screens from computerized equipment, generating visual descriptions of candidates' screens based on at least one of the visual descriptions of outlines of candidates' screens and the screen content data of candidates' screens, followed by identifying the screen pointed at by UPI device **100** based on the estimated orientation of UPI device **100**, on the estimated azimuth of UPI device **100**, on the generated frontal visual representation for UPI device **100** and on at least one of the obtained location parameters of candidates' screens and the generate visual descriptions of candidates' screens, as described in the discussion of FIG. **4** of U.S. patent application Ser. No. 16/931,391. It should be emphasized that the use of the term “obtaining features from features database **250**” should be viewed as a general step of obtaining all necessary data about the locations, the orientations, the physical dimensions, the outlines and the content of the candidates' screens. For example, the locations and orientation of fixed screens (e.g., TV screen **620** or presentation screen **640**) may be known and stored in features database **250**, while the locations and orientations of handheld devises may be constantly calculated and updated, employing the same approach described for UPI device **100** and utilizing the images captured by the rear-facing cameras of the handheld devices. Note, of course, that this approach for calculating and updating the locations and orientations of handheld devises is described in several of the references cited in U.S. patent application Ser. No. 16/931,391. Moreover, as discussed for some objects of interest in FIG. **7** of U.S. patent application Ser. No. 16/931,391, as practically all computerized equipment incorporates wireless communication elements, directional and strength measurements of wireless signals may be used in the identification process. For example, assuming that a particular UPI device **100** can be used interchangeably with several different handheld phones, using measurements of the WiFi signals strength (or the strength of any other wireless signals) from these different phones may be used to assist in the determination of which of the different handheld phones with its screen is pointed at by that UPI device **100**. In addition, if a particular UPI device **100** is used entirely with a particular screen, for example, the single handheld device used the user of that UPI device **100** with its screen, the process of identifying the screen described above may be completely bypassed and eliminated.

[0042] Once a screen is identified, user **200** may further use UPI device **100** to activate the identified screen. The key to activating the identified screen is the estimation of the relative spatial relations between UPI device **100** and the screen, and in particular the relative spatial relations between UPI device **100** and a handheld device with a screen. Using the relative spatial relations between UPI device **100** and the screen it is possible to estimate the screen coordinates at the location pointed at by tip **102** of UPI device **100**, which are the coordinates relative to one corner of the screen where the extension of the main axis of UPI device **100** in the frontal direction from tip **102** of UPI device **100**, represented by pointing ray **105**, meets the identified screen. The estimating of the relative spatial relations between UPI device **100** and the screen, as well as the estimation of the screen coordinates at the location pointed at by tip **102** of UPI device **100**, may use the estimated orientation and azimuth of UPI device **100** and the known or the estimated orientation and azimuth of the identified screen. The estimation may also use the information about the obtained visual descriptions of outline of the identified screen and the content of the identified screen for match filtering with the image captured by camera **110** of between UPI device **100** for match filtering to establish the exact spatial relations between UPI device **100** and the identified screen. It is important to note that the estimated orientation and azimuth of UPI device **100** and the known or the estimated orientation and azimuth of the identified screen may be used to generate modified visual descriptions of outline of the identified screen and the content of the identified screen, using spatial translation, rotation and perspective projection of the obtained visual descriptions of outline of the identified screen and the content of the identified screen, to compensate for the resulted spatial shift, angle and perspective due to the relative locations and angles between UPI device **100** and the identified screen. Moreover, if the screen is a handheld device, such as a smartphone or tablet, it is possible to use images captured by either the front-facing camera or the back-facing camera of the handheld device to generate visual representations that include the outline of UPI device **100** and to use these visual representations for the estimating of the relative spatial relations between UPI device **100** and the handheld device and its screen, as well as the estimation of the screen coordinates at the location pointed at by tip **102** of UPI device **100**. In addition, LIDAR **115**, as well as the equivalent components on the handheld device or other screen, may be used for the estimating of the relative spatial relations between UPI device **100** and the screen, as well as the estimation of the screen coordinates at the location pointed at by tip **102** of UPI device **100**.

[0043] The identified screen may be activated by user **200** of UPI device **100**, such as selecting items on the identified screen, opening a hyperlink, scrolling, zooming, highlighting or any other activation commonly performed by a mouse or other common pointing element, based on the estimated screen coordinates at the location pointed at by tip **102** of UPI device **100** and based on activating signals from UPI device **100** provided by at least one of accelerometer **122**, magnetometer **124**, camera **110**, gyroscope **126**, tactile sensors **132**, microphone **134** and fingerprint detection sensor **136**, as described in step **760**. In particular, the 2-dimensional activations based on the screen coordinates at the location pointed at by tip **102** of UPI device **100**, which is similar to activation by a mouse or other common pointing elements, may be extended to 3-dimensional activations. It is possible to trace the relative 3-dimensional motion between UPI device **100** and the identified screen and to activate the identified screen based on that 3-dimensional motion, such as zooming in and out based on the changing distance between UPI device **100** and the identified screen, which may be detected by any of user location sensing components **120**, camera **110** and LIDAR **115**, as well as the equivalent components on the selected screen, such as the location/motion sensing components, the front-facing camera, the rear-facing camera and LIDAR sensors of the selected screen.

[0044] The interaction of UPI device **100** with the screen of handheld device **830** (which may play the role of handheld device **205**) is of particular interest, as handheld device **830** may not be stationary and therefore the screen coordinates for the screen of handheld device **830** at the location pointed at by tip **102** of UPI device **100** may also depend on the motion of handheld device **830**. One aspect is that the relative 3-dimensional motion between UPI device **100** and handheld device **830** may be further tracked by the motion sensors of handheld device **830**. Another aspect is that since handheld device **830** includes a front-facing camera and a rear-facing camera, for handheld device **830** the estimation of the screen coordinates at the location pointed at by tip **102** of UPI device **100** and the activation of the screen may also be based on at least one image taken by at least one of the front-facing camera and the rear-facing camera of handheld device **830**. When UPI device **100** points to the screen of handheld device **830**, image of UPI device **100** may be captured by the front-facing camera, whereas when UPI device **100** points to identified objects of interest in the field of view of the rear-facing camera (depicted by FIG. **8** and discussed below), image of UPI device **100** may be captured by the rear-facing camera. An image of UPI device **100** captured by any of the front-facing camera and the rear-facing camera may be used to further assist in step **760** in the estimation of the screen coordinates for the screen of handheld device **830** at the location pointed at by tip **102** of UPI device **100**. An image of UPI device **100** captured by any of the front-facing camera and the rear-facing camera may also be used in the estimation of the relative positions of UPI device **100** and handheld device **830** even if the tip **102** of UPI device **100** is not pointing at the screen of handheld device **830**. Moreover, particular shapes and color marking on UPI device **100** may further improve the accuracy and efficiency of the estimation of the relative position between UPI device **100** and handheld device **830**, using images captured by any camera on handheld device **830**. In addition, measurements by LIDAR **115** or by a LIDAR of handheld device **830** of the distance between UPI device **100** and handheld device **830** may be used in tracking the relative position of UPI device **100** with respect to handheld device **830**. Moreover, a multi-beams LIDAR of handheld device **830** may be used to provide very accurate estimation of the relative spatial relations between UPI device **100** and handheld device **830** and its screen, as well as the estimation of the screen coordinates at the location pointed at by tip **102** of UPI device **100**.

[0045] Handheld device **830** and UPI device **100** may be configured to provide augmented reality (AR) window into the environment, providing orientation and entertainment to user **200**. Since handheld device **830** includes a rear-facing camera, UPI device **100** may be used to point at objects viewed on the screen of handheld device **830** that are either captured by the rear-facing camera or are virtually generated on the screen of handheld device **830** as augmented reality objects in the environment captured by the rear-facing camera. FIG. **8** shows several examples of UPI device **100** interacting with real and virtual objects that are displayed on the screen of handheld device **830**. The first example is of pointing at real objects of interest in the environment such as building **826** and at public transportation vehicle **822**. As discussed in U.S. patent application Ser. No. 16/931,391, using location information using GPS or other positioning systems and improving the accuracy of the location and directional information, the pointing at building **826** may provide user **200** with information such as businesses and residences in building **826**, and the pointing at transportation vehicle **822** may provide information such as schedule and destinations of transportation vehicle **822**. Using UPI device **100**, it is possible to generate markers that point or interact with the images of the real objects of interest. In this first example, the images of these real objects of interest are captured by the rear-facing camera and are displayed on the screen of handheld device **830**. As user **200** points to these real objects of interest, interacting marker **810** extends pointing ray **105** on the screen of handheld device **830**. For the element of building **826**, UPI device **100** is above the plane of the screen and interacting marker **810** starts on the screen at the point where pointing ray **105** "reaches" the screen of handheld device **830** and ends near the image of building **826** on the screen. For public transportation vehicle **822**, UPI device **100** is below the plane of the screen and therefore its image (part or whole) may be captured by the rear-facing camera of handheld device **830**. In this example, interacting marker **810** starts on the screen at the image of tip **102** of UPI device **100** and ends near the image of public transportation vehicle **822** on the screen. A second example is of augmented reality games that may place virtual objects **820** overlaid in the environment in the content of the image or video captured by the rear-facing camera of handheld device **830**. As the locations on the screen of handheld device **830** of such of the game virtual object are determined by the game design and are known, UPI device **100** may be used to point to the screen of handheld device **830** and interact, using interacting content, with virtual objects **820** via the screen of handheld device **830**. In particular, interaction graphic content may be generated on the screen of handheld device **830** for the interacting with virtual objects **820**, such as a lightning bolt to fight an evil wizard, a ballistic path to throw a ball to hit a magical creature, or a lasso to catch a flying dragon. The interacting with real objects of interest in the environment and with virtual objects generated for games may be also carried out by providing output by UPI device **100** with its LED, its vibration motor, its loudspeaker or its optional screen, as well as providing output by handheld device **830** with its loudspeaker or its vibration motor.

Claims

1. A method of providing information of an object in the surroundings using a universal pointing and interacting (UPI) device and a handheld device, the method comprises: measuring acceleration parameters of the UPI device by an accelerometer; measuring magnetic field parameters of the UPI device by a magnetometer; capturing at least one image by a camera positioned at a tip of the UPI device; processing by a computation component the measured acceleration parameters to estimate an orientation of the UPI device, the measured magnetic field parameters to estimate an azimuth of the UPI device and the at least one captured image to generate a frontal visual representation for the UPI device; transmitting by a communication component the estimated orientation of the UPI device, the estimated azimuth of the UPI device and the generated frontal visual representation for the UPI device from the UPI device to the handheld device; estimating location parameters of the handheld device using at least one of GPS signals, cellular signals, Wi-Fi signals, Bluetooth signals and dedicated wireless beacons signals; obtaining features of candidate objects in the surroundings from a features database; identifying the object in the surroundings from the candidate objects in the surroundings based on the estimated orientation of the UPI device, on the estimated azimuth of the UPI device, on the generated frontal visual representation for the UPI device, on the estimated location parameters of the handheld device and on the obtained features of candidates objects in the surroundings. providing information about the identified object in the surrounding by at least one of a screen of the handheld device, a loudspeaker of the handheld device, an earpiece connected to the handheld device and a headphone connected to the handheld device.
2. The method of claim 1, wherein the identified object is a building and the information about the identified object is the address of the building.
3. The method of claim 1, wherein the identified object is a commercial business and the information about the identified object is at least one of a type of commercial business and opening hours of the commercial business.
4. The method of claim 1, wherein the identified object is a restaurant and the information about the identified object is at least one of opening hours of the restaurant and menu of the restaurant.

5. The method of claim 1, wherein the identified object is a public sign that includes at least one of alphanumerical information and image information and the provided information about the identified object is at least one of the alphanumerical information and the image information.